

# Stroke Prognosis & Clinical Scores

Clinical prediction scales for ischemic and hemorrhagic stroke outcomes.

## STROKE PROGNOSIS SCALES

Ischemic Stroke: ASTRAL & PLAN

Hemorrhagic Stroke: ICH Score

### ASTRAL Score (Acute Ischemic Stroke)

Predicts 90-day poor functional outcome (mRS > 2)

| Predictor Variable                  | Points            |
|-------------------------------------|-------------------|
| Age                                 | 1 pt per 5 years  |
| Severity (NIHSS)                    | 1 pt per NIHSS pt |
| Time to admission >3h               | 2 pts             |
| Range of visual fields (defect)     | 2 pts             |
| Acute glucose (<3.7 or >7.3 mmol/L) | 1 pt              |
| Level of consciousness (impaired)   | 3 pts             |

#### Score vs. 90-Day Unfavorable Outcome (mRS > 2) Risk:

|     |      |
|-----|------|
| <20 | <5%  |
| 23  | 15%  |
| 31  | 50%  |
| 35  | 70%  |
| 40+ | >90% |

# PLAN Score (Acute Ischemic Stroke)

Bedside prediction of 30-day mortality/dependence

| Predictor Domain & Variables           | Points                   |
|----------------------------------------|--------------------------|
| Preadmission dependence / Cancer       | 1.5 pts each             |
| Preadmission CHF / Atrial Fibrillation | 1.0 pt each              |
| Level of consciousness (reduced)       | 5.0 pts                  |
| Age (decades)                          | 1 pt per decade (max 10) |
| Neurologic: Leg / Arm weakness         | 2 pts each               |
| Neurologic: Aphasia or Neglect         | 1.0 pt                   |

## Score vs. 30-Day Mortality Risk:

|      |      |
|------|------|
| <6   | 0.7% |
| 9-10 | 4.4% |
| 13   | 15%  |
| 16   | 35%  |
| >19  | >65% |

# ICH Score (Intracerebral Hemorrhage)

Predicts 30-day mortality in spontaneous ICH

| Predictor Component                                  | Points  |
|------------------------------------------------------|---------|
| GCS Score: 3–4 (2 pts)   5–12 (1 pt)   13–15 (0 pts) | 0–2 pts |
| Age: ≥ 80 years                                      | 1 pt    |
| ICH Volume: ≥ 30 mL                                  | 1 pt    |
| Intraventricular Hemorrhage (IVH): Present           | 1 pt    |
| Infratentorial Origin of Hemorrhage                  | 1 pt    |

## Score vs. 30-Day Mortality Risk:

|   |      |
|---|------|
| 0 | 0%   |
| 1 | 13%  |
| 2 | 26%  |
| 3 | 72%  |
| 4 | 97%  |
| 5 | 100% |

Historical cohort rates, not individual predictions. No percentage is assigned here to score 6; do not extrapolate a certain fatal outcome.

## Modified Rankin Scale (mRS)

*The gold standard for assessing global functional recovery*

### Grade Clinical Description of Recovery State

- 0 No symptoms at all.
- 1 No significant disability despite symptoms; able to carry out all usual duties.
- 2 Slight disability; unable to carry out all previous activities but **independent**.
- 3 Moderate disability; requires some help but **able to walk unassisted**.
- 4 Moderately severe; **unable to walk or attend to bodily needs** without assistance.
- 5 Severe disability; bedridden, incontinent, requiring constant nursing care.
- 6 Dead.

### Prognostication Principles & Limitations

- **Not for Care Limitations:** These clinical scores serve to quantify severity, improve inter-provider communication, and assist in counseling. They **MUST NOT** be used in isolation as the sole basis for withholding reperfusion therapies, surgical decompression, or withdrawing life-sustaining treatment (avoiding the self-fulfilling prophecy of poor outcome).
- **Dynamic Evaluation:** Clinical trajectory over the first 24–72 hours is often more predictive of final recovery than any single point-in-time calculation upon hospital admission.
- **Acute ICH — Hemostatic Therapy:** The ICH Score is prognostic, not a treatment target. Early intensive blood-pressure lowering and hematoma-directed care remain the evidence-based acute levers (2022 AHA/ASA ICH guideline). Recombinant factor VIIa (rFVIIa) given within 2h slowed hematoma growth but did **not** improve 180-day function and increased thromboembolic events (FASTEST, 2026; PMID 41653933) — **not** recommended for routine use.

**ASTRAL Score:** Ntaios G, et al. *Neurology*. 2012;78(24):1916-22. [PMID: 22649218](#)

**PLAN Score:** O'Donnell MJ, et al. *Arch Intern Med*. 2012;172(20):1548-56. [PMID: 23147454](#)

**ICH Score:** Hemphill JC 3rd, et al. *Stroke*. 2001;32:891-7. [PMID: 11283388](#)

**mRS Scale:** van Swieten JC, et al. *Stroke*. 1988;19:604-7. [PMID: 3363593](#)